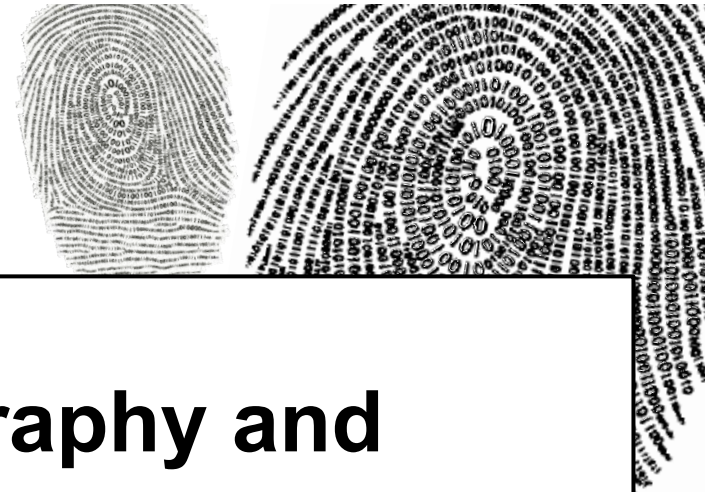
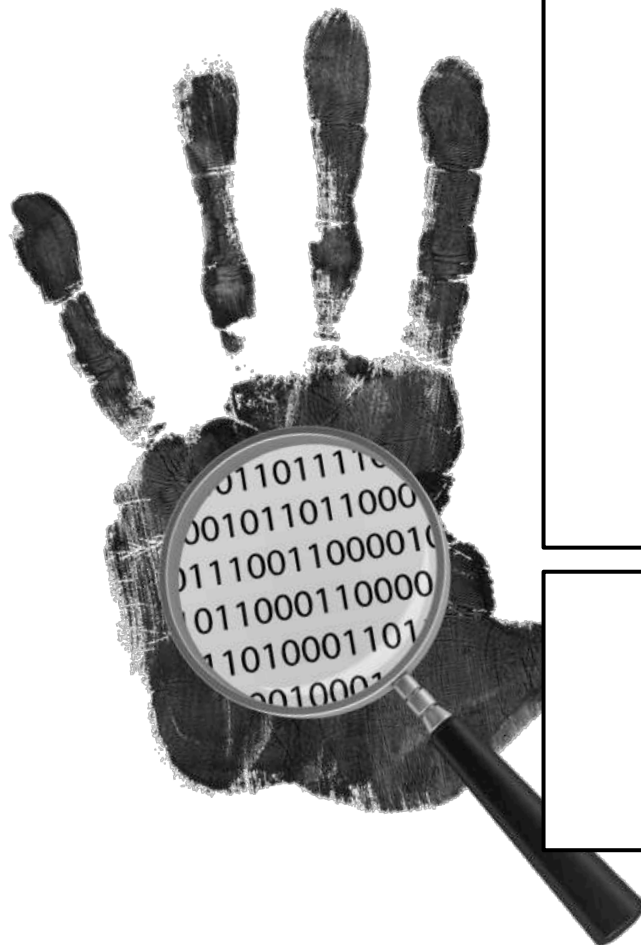




Steganography and Watermarking

Section II. Techniques and Tools for Computer Forensics



CSF: Forensics Cyber-Security

Fall 2017

Nuno Santos



Previous class





Can you spot a difference between these images?



Image A



Image B



Do they carry the same amount of information?

- ▶ No! Image B hides a secretly encoded message



Image B

decode

Attack at 14:00!

Hidden message



Steganography

- ▶ Used for concealment of communications in various crimes, e.g., terrorism, botnet management, data exfiltration, etc.

NEWS

Criminals using steganography tricks to manage ZBOT attacks

Researchers at Trend Micro have discovered examples of botnets being managed by steganography techniques.

Hidden file upload

Hackers used data exfiltration based on video steganography

November 29, 2014 By [Pierluigi Paganini](#)

G+1 28

f My Page f Like 131

Security experts have detected an attack against a major firm that used a data exfiltration technique based on the video steganography.

Hidden file download

Steganography and terrorism: Why ISIS relies on it so much

Written by: [Vicky Nanjappa](#) Published: Monday, March 2, 2015, 11:17 [IST]

Hidden bidirectional communication



Class roadmap

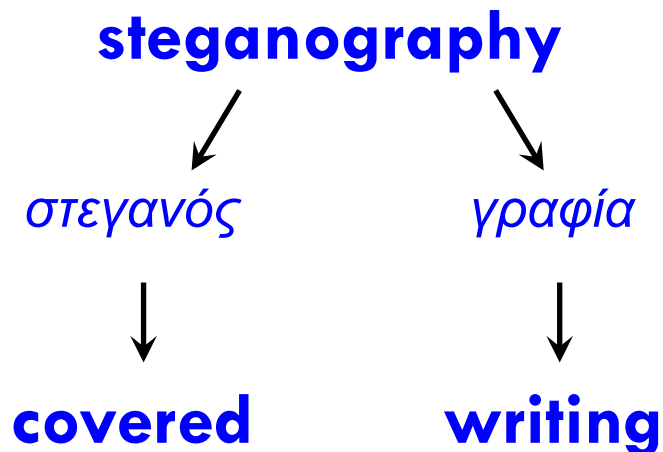
- ▶ Steganography
- ▶ Steganalysis
- ▶ Watermarking

Steganography



Steganography defined

- ▶ **Steganography:** Art and science of communicating in a way that hides the existence of a message
 - ▶ From the Greek words *steganos* and *graphy*



- ▶ Steganography simply takes one piece of information (**secret**) and hides it within another (**carrier / cover**)



Cryptography vs. steganography

- ▶ **Cryptography**

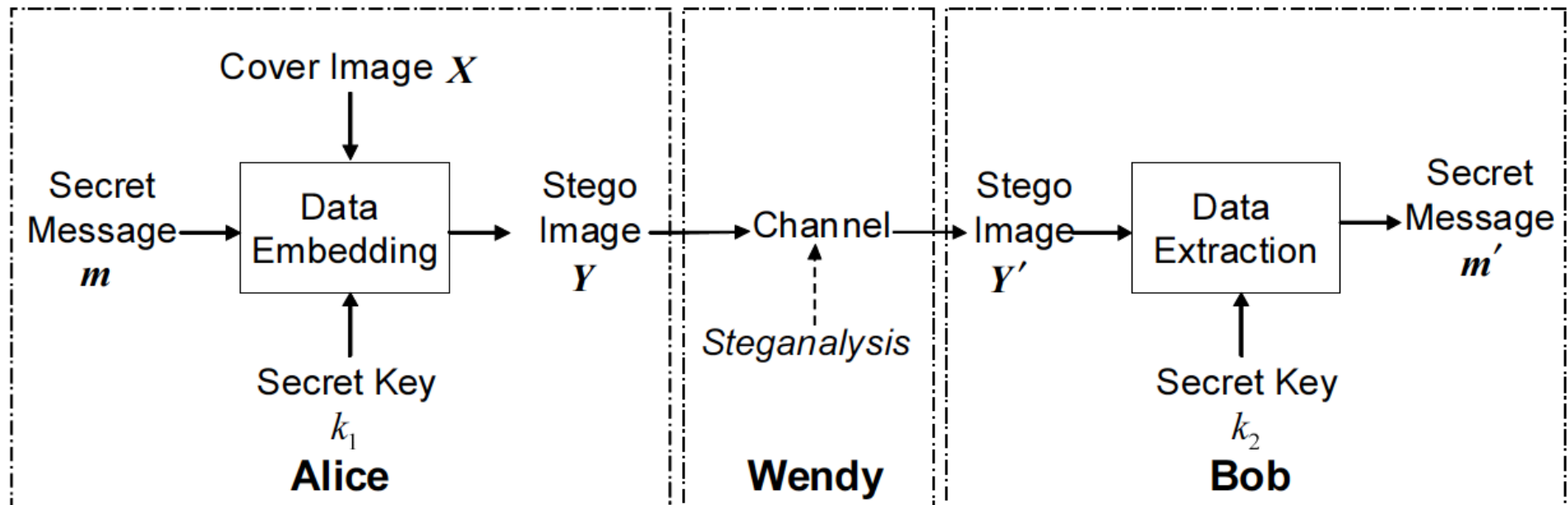
- ▶ Is about protecting the content of messages (their meaning)

- ▶ **Steganography**

- ▶ Is about concealing the existence of messages



Model of a steganography system



Wendy can be **active** (makes $Y = Y'$) or **passive** (is Y suspicious?)



Desired properties of a steganography

- ▶ **Imperceptibility**

- ▶ The resultant stego image appears innocuous enough, and not having severe visual artifacts

- ▶ **Security**

- ▶ Must be able to resist active as well as passive attacks

- ▶ **Capacity**

- ▶ The hiding capacity should be as high as possible



Early steganography in Ancient Greece: Tattoos

- ▶ In the 5th century BC, Histaiacus shaved a slave's head, tattooed a message on his skull and the slave was dispatched with the message after his hair grew back
 - ▶ He wanted to instigate revolt against Persians



Today, planning the escape:
tattoo contains hidden
blueprints of Fox River
State Penitentiary



In Ancient Rome: Invisible ink

- ▶ Ancient Romans used to write between lines using **invisible ink**
 - ▶ Based on various natural substances such as fruit juices, urine, and milk
 - ▶ Messages appear only when heated



Using lemon



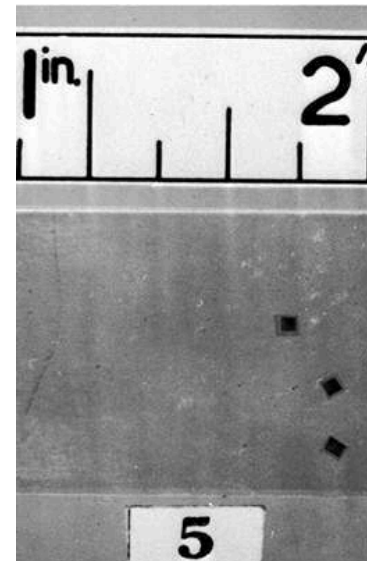
The XXI century way: UV pen



Using milk



- ▶ 14





Another example from the WWs: Null-Cipher

- ▶ Null cipher: plaintext is mixed with a large amount of non-cipher material (termed null characters)
- ▶ Message sent by a German spy during World war-I:

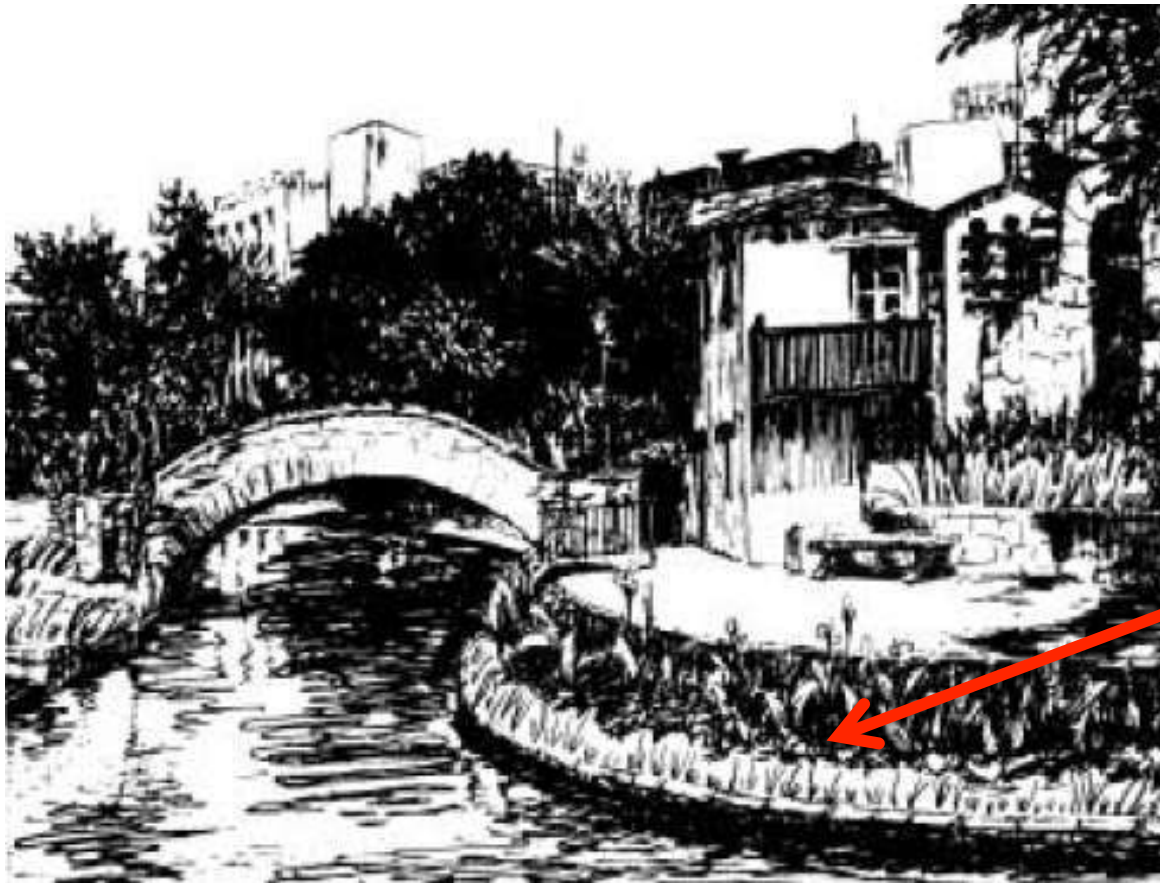
PRESIDENT' S EMBARGO RULING SHOULD HAVE IMMEDIATE
NOTICE. GRAVE SITUATION AFFECTING INTERNATIONAL LAW.
STATEMENT FORESHADOWS RUIN OF MANY NEURALS.
YELLOW JOURNALS UNIFYING NATIONAL EXCITEMENT
IMMENSELY.

Pershing sails from NY June 1



Ideas from modern times: Drawings

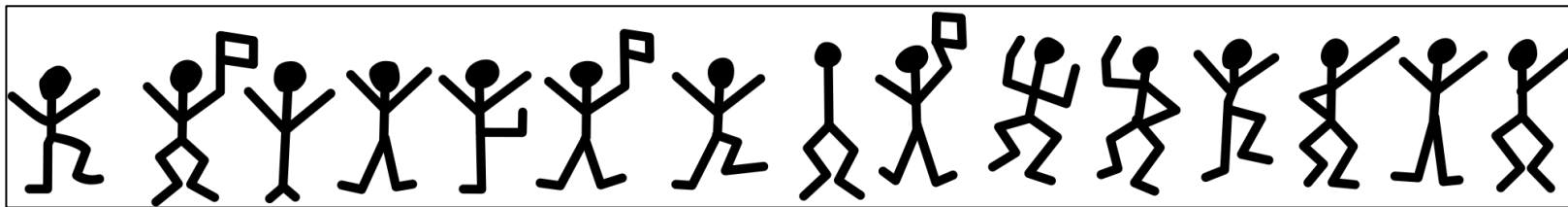
- ▶ In 1945, Morse code was concealed in a drawing
 - ▶ Hidden information is encoded onto the grass length alongside the river





More drawings: Pictographs

- ▶ Secret message hidden in an apparently innocuous sequence of pictographs
- ▶ In the short story of Sherlock Holmes 'The Adventures of the Dancing Men' a man tells Holmes that his wife, Elsie, receives notes with dancing men on them





More drawings: Pictographs

- ▶ Dancing men turned out to be a secret code
 - ▶ Men with a flag denote the last letter of a word

A	B	C	D	E	F	G	H	I	J	K	L	M
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	0			



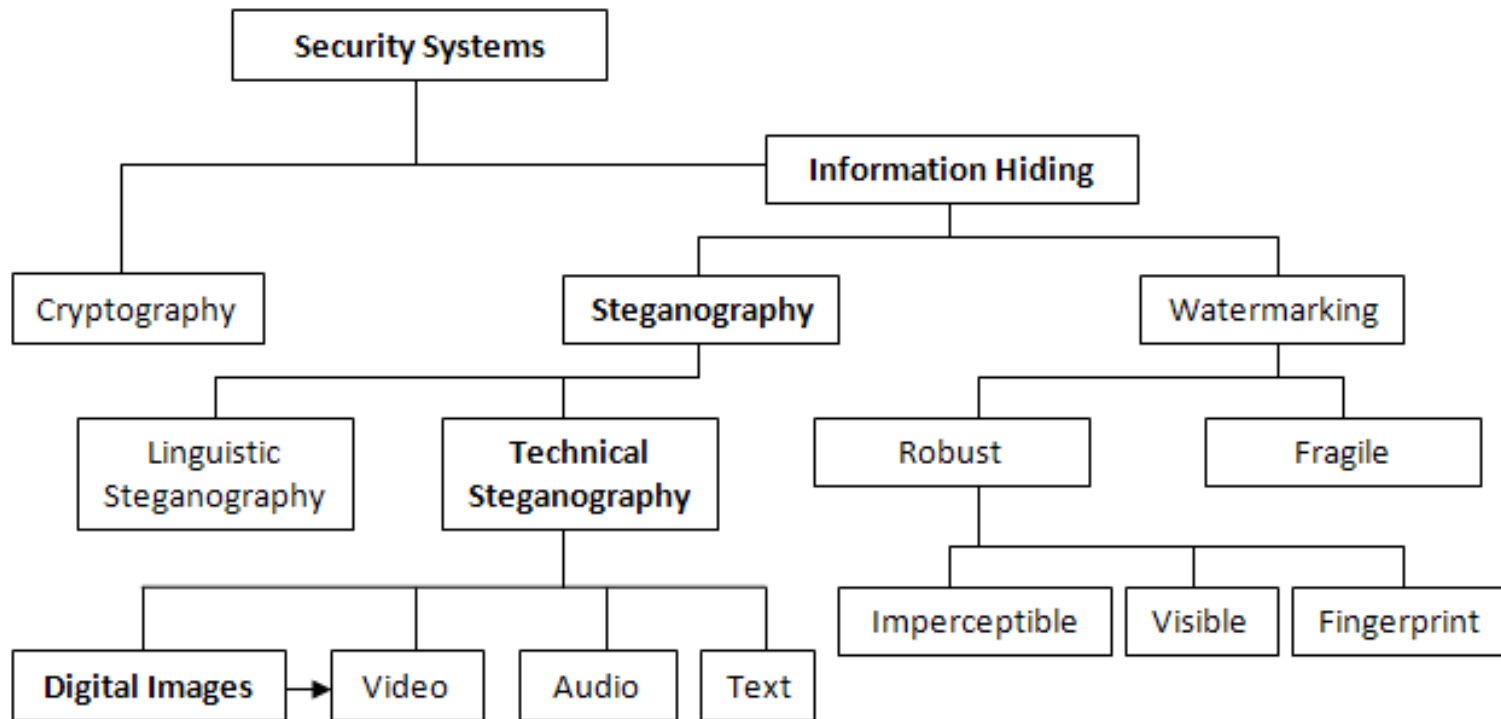
More old ideas

- ▶ Pinpricks in maps
- ▶ Dotted I's and crossed T's
- ▶ Deliberate misspellings or errors, e.g., errors in trivia books, etc
- ▶ Unusual languages: e.g., navajo, peculiar sounds used esp., in Guerilla warfare



Digital steganography

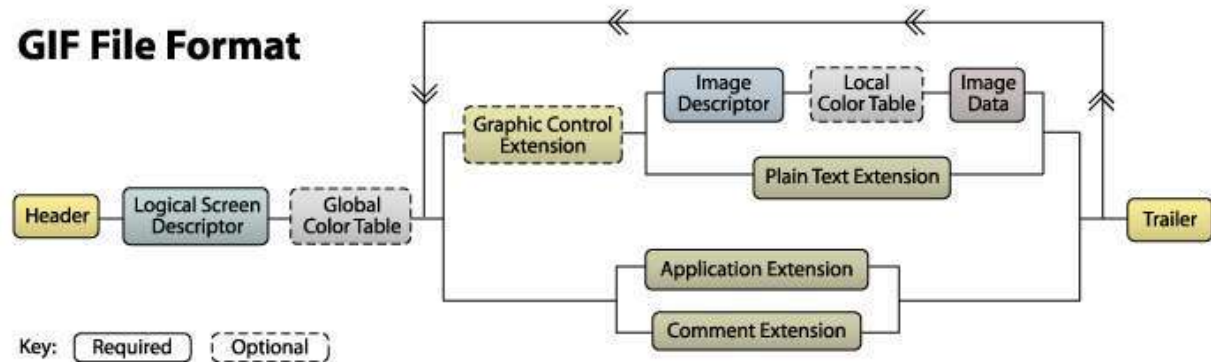
- ▶ **Digital steganography** works by encoding secret bits in files, such as photos or audio files, with secret data
 - ▶ The secret message and the carrier message are digital objects





Another example: Append secret into headers

- ▶ **Extended File Information (EXIF)**
 - ▶ Standard used by digital camera manufacturers to store information in the image file, such as, the make and model of a camera, the time of picture, etc.
- ▶ **GIF's comment extension**

[illegible]

the inserted text string highlighted in a box



EXPLOITING THE SPACE DOMAIN

- ▶ A steganographer modifies the secret data and the cover medium in the spatial domain, which involves encoding at the level of the image pixels



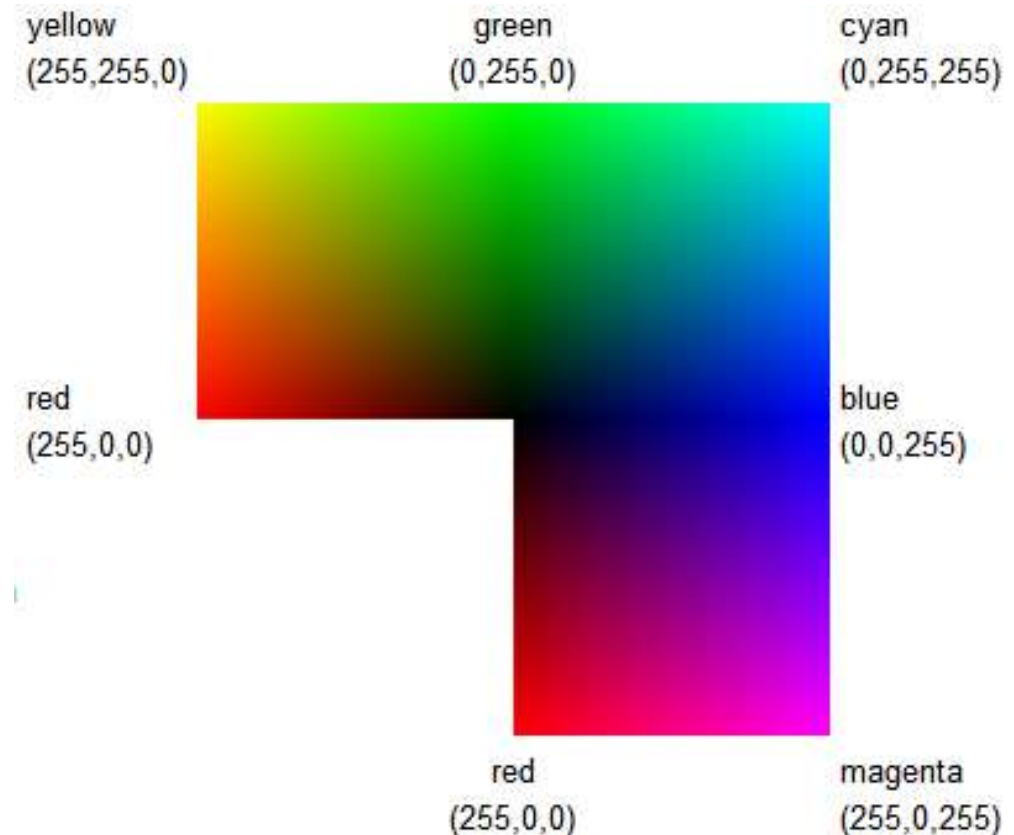
Pixel



► 24-bit RGB image files

- Each pixel encoded by 3 byte values for **red**, **green**, and **blue**

(0, 0, 0) is black
 (255, 255, 255) is white
 (255, 0, 0) is red
 (0, 255, 0) is green
 (0, 0, 255) is blue
 (255, 255, 0) is yellow
 (0, 255, 255) is cyan
 (255, 0, 255) is magenta





Grayscale image

- ▶ Pixels encode intensities in the form of shade of gray:



from black (lowest value) to white (largest value)

- ▶ Encoding defines how many bits per pixel: 8, 16, 32...
 - ▶ More bits means more levels of shade





A common digital steganography technique: LSB

► Least Significant Bit (LSB)

- The LSB of a pixel is used to encode hidden information

- Example: Suppose we want to encode the letter A in the following 8 bytes of a **8-bit grayscale carrier image**

- “A” → ASCII 65 or binary 01000001

01011101	11010000	00011100	10101100
11100111	10000111	01101011	11100011

becomes

0101110 <u>0</u>	1101000 <u>1</u>	0001110 <u>0</u>	1010110 <u>0</u>
1110011 <u>0</u>	1000011 <u>0</u>	0110101 <u>0</u>	1110001 <u>1</u>



What happens to the image?



Cover



Stego (embedding secret 'A')

► This works because:

1. Digital image or sound files can be altered to a certain extent without losing their functionality
2. Humans are unable to distinguish minor changes in image color or sound quality



Leveraging the RGB channels in color images

- ▶ Consider a 24 bit picture
- ▶ Data to be inserted: character 'A': (01000001)
- ▶ Host pixels: 3 pixel used to store one character of 8-bits

R	G	B
00100111	11101001	11001000
00100111	11001000	11101001
11001000	00100111	11101001

Embedding 'A'

0010011 <u>0</u>	1110100 <u>1</u>	1100100 <u>0</u>
0010011 <u>0</u>	1100100 <u>0</u>	1110100 <u>0</u>
1100100 <u>0</u>	0010011 <u>1</u>	11101001



Apply to LSB to color image carrier

- ▶ Tweaking the LSB is only a small change in image color
 - ▶ Pixel = 00100111, 11101001, 11001000
 - ▶ Pixel' = 00100110, 11101001, 11001000



Original image



LSB modified to hide info

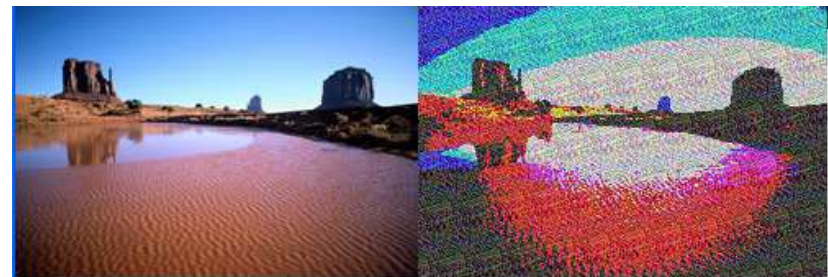


It's possible to use different bits for encoding

- ▶ Different results in terms of capacity and added noise
 - ▶ More bits means higher capacity, but higher noise
 - ▶ Emerges a side effect named banding



4 LSB modified produces banding



7 bits



6 bits

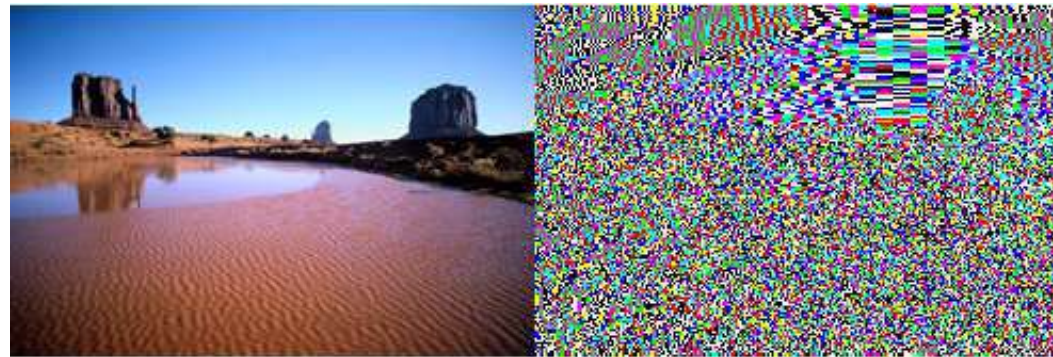


All 8 bits



What if we change the most significant bit?

- ▶ Here's the result:



Bit 8 vs. Bit 1

- ▶ Why is it so?



What kind of data can be used as payload?

- ▶ An arbitrary sequence of binary data
 - ▶ Namely, text or another image
- ▶ You can add encrypted data too
- ▶ As more pixels are used, chances of detection increase
 - ▶ According to researchers on an average only 50% of the pixels actually change from 0-1 or 1-0





LSB embedding and extraction algorithms

- ▶ Any cover can be represented by a sequence of numbers c_i of length $l(c)$ (i.e., $1 \leq i \leq l(c)$)
- ▶ We denote the stego object by s which is again a sequence s_i of length $l(c)$
- ▶ The secret message is m , length of m by $l(m)$, and the bits forming m by m_i , $1 \leq i \leq l(m)$

Algorithm 3.1 Embedding process: least significant bit substitution

```
for  $i = 1, \dots, \ell(c)$  do
     $s_i \leftarrow c_i$ 
end for
for  $i = 1 \dots, \ell(m)$  do
    compute index  $j_i$  where to store  $i$ th message bit
     $s_{j_i} \leftarrow c_{j_i} \oplus m_i$ 
end for
```

Algorithm 3.2 Extraction process: least significant bit substitution

```
for  $i = 1, \dots, \ell(M)$  do
    compute index  $j_i$  where the  $i$ th message bit is stored
     $m_i \leftarrow \text{LSB}(c_{j_i})$ 
end for
```



Adding a key protection

- ▶ Make it not obvious where to find the stego pixels containing the secret bits
- ▶ **Shifting:** Simple approach that indicates an index $\neq 0$ of the first stego pixel to embed bit secrets
 - ▶ Example: $\text{index_first_stego_pixel} = \text{key} \% 27$
 - ▶ The modulo operator ensures the shift is bound
- ▶ **Random interval:** the use of a pseudorandom number generator to spread the secret message over the cover



Random interval method

If both communication partners share a stego-key k usable as a seed for a random number generator, they can create a random sequence $k_1, \dots, k_{l(m)}$ and use the elements with indices

- ▶ The distance between two embedded bits is pseudorandom

$$\begin{aligned} j_1 &= k_1 \\ j_i &= j_{i-1} + k_i, \quad i \geq 2 \end{aligned}$$

j_i indicates next index

Algorithm 3.3 Embedding process: random interval method

```
for  $i = 1 \dots, \ell(c)$  do
     $s_i \leftarrow c_i$ 
end for
generate random sequence  $k_i$  using seed  $k$ 
 $n \leftarrow k_1$ 
for  $i = 1, \dots, \ell(m)$  do
     $s_n \leftarrow c_n \oplus m_i$ 
     $n \leftarrow n + k_i$ 
end for
```

Algorithm 3.4 Extraction process: random interval method

```
generate random sequence  $k_i$  using seed  $k$ 
 $n \leftarrow k_1$ 
for  $i = 1, \dots, \ell(m)$  do
     $m_i \leftarrow \text{LSB}(c_n)$ 
     $n \leftarrow n + k_i$ 
end for
```



Pixels of a carrier image to be used

- ▶ As more pixels are used, chances of detection increase
 - ▶ According to researchers on an average only 50% of the pixels actually change from 0-1 or 1-0
- ▶ Select the pixels for holding the data on the basis of a **key** which can be a random number
 - ▶ The key serves as seed to a random number generator



LSB: The good, the bad, and the ugly

▶ **The good**

- ▶ Simple to implement
- ▶ Allows for large payload: $\text{Max payload} = b * p$
 - ▶ b = number of bytes per pixel, p = number of pixels of cover image

▶ **The bad**

- ▶ Easy to figure out message if attacker knows the msg is there
 - ▶ Vulnerable to statistical analysis

▶ **The ugly**

- ▶ Integrity is extremely frail
- ▶ Easy for attacker to corrupt the message
 - ▶ E.g., just randomize the LSBs himself
- ▶ Vulnerable to unintentional corruption
 - ▶ E.g., image cropping, conversion to jpeg and back, etc



Digital steganography techniques

▶ **Substitution methods**

- ▶ Substitute redundant parts of a cover with a secret message
- ▶ Bit plane methods (LSB), palette-based methods

▶ **Transform method techniques**

- ▶ Embed secret info in a transform space of a signal (e.g., frequency domain)

▶ **Distortion techniques**

- ▶ Store information by signal distortion and measure the deviation from the original cover in the decoding step

▶ **Cover generation methods**

- ▶ Encode information by creating a cover object (e.g., fractal generation)



Steganography tools

- ▶ Steganos
- ▶ S-Tools (GIF, JPEG)
- ▶ StegHide (WAV, BMP)
- ▶ Invisible Secrets (JPEG)
- ▶ JPHide
- ▶ Camouflage
- ▶ Hiderman
- ▶ Many others...

Steganalysis



“Steganography messages are difficult to detect by investigators”

Detection is difficult

For an investigating officer detecting steganography is a nightmare. There is absolutely no record to show that the sender and the receiver had ever communicated. They do not exchange calls or emails.

What investigating agencies have been doing is keeping a track of all downloaded pictures on the web. Pictures that are downloaded in places where the terrorist networks are strong are part of the data base. They would then keep a watch on these pictures closely to see if there are messages coded into them.

<http://www.oneindia.com/feature/steganography-and-terrorism-why-isis-relies-on-it-so-much-1670728.html>



What is Steganalysis ?

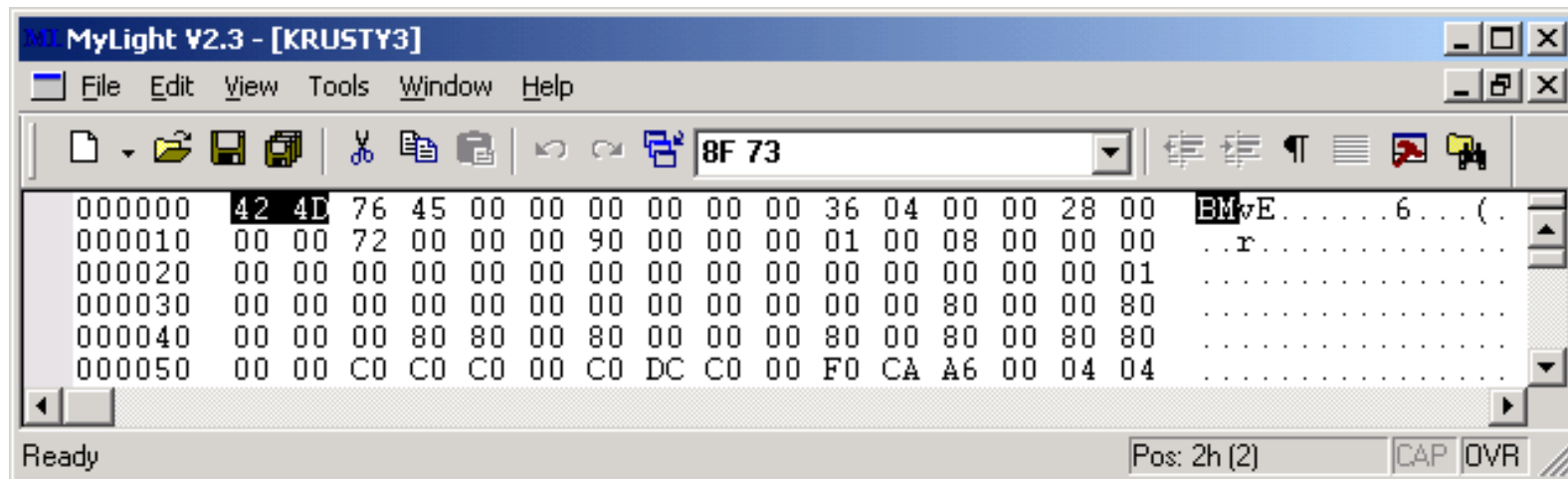
- ▶ The art and science of detecting hidden data
 - ▶ By identifying the existence of a hidden message, perhaps we can identify the tools used to hide it
 - ▶ If we identify the tool, perhaps we can use that tool to extract the original message

- ▶ Why is it important?
 - ▶ Prevent terrorist attacks
 - ▶ Catch people engaging in illegal activities
 - ▶ Discourage piracy



Case study: Hiderman

- ▶ Let's examine a slightly sophisticated stego program – Hiderman
- ▶ After hiding a message with Hiderman, review the file with favorite Hex Tool
 - ▶ Viewing the Header information (beginning of the file) we see that it's a Bitmap as indicated by the "BM" file signature





Case study: Hiderman

- ▶ View the end of the file, comparing the virgin file to the carrier file
 - ▶ Note the data appended to the file

```
004520 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01
004530 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01
004540 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01
004550 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01
004560 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01
004570 01 01 01 01 01 00 00
```

3 byte(s) selected. Pos: 4573h (17779) CAP OVR

```
004560 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01
004570 01 01 01 01 00 00 8F 73 08 00 8E 71 00 00 A7 59
004580 36 7A E5 1A 7A 7A EE 11 7A 66 E5 1F 65 2E FA 09
004590 74 7A ED 12 7A 7A 7A 7A 7A 03 00 76 45 00 00 00
0045a0 00 03 00 23 00 00 00 00 00 02 09 01 01 02 02 07
0045b0 06 09 01 01 02 02 05 01 04 02 09 12 43 44 4E
```

Quick find combo Ctrl+Q Pos: 4576h (17782) CAP OVR



Case study: Hiderman

- ▶ In addition, note last three characters “CDN”: 43 44 4E in HEX

```
MyLight V2.3 - [KRUSTY3]
File Edit View Tools Window Help
8F 73
004560 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01 01 .....
004570 01 01 01 01 00 00 8F 73 08 00 8E 71 00 00 A7 59 .....s...q...Y
004580 36 7A E5 1A 7A 7A EE 11 7A 66 E5 1F 65 2E FA 09 6z..zz..zf..e...
004590 74 7A ED 12 7A 7A 7A 7A 7A 03 00 76 45 00 00 00 tz..zzzzzz..vE...
0045a0 00 03 00 23 00 00 00 00 00 02 09 01 01 02 02 07 ...#.....
0045b0 06 09 01 01 02 02 05 01 04 02 09 12 43 44 4E .....CDN
Quick find combo Ctrl+Q Pos: 45BCh (17852) CAP OVR
```

- ▶ Hiding different messages in different files with different passwords, (“CDN”) always appended: Signature found!

```
MyLight V2.3 - [cartman]
File Edit View Tools Window Help
8F 73
003890 24 24 24 24 24 24 24 24 24 24 24 24 24 24 24 24 $$$$$$$$$$$$$$$$
0038a0 24 24 24 24 24 24 24 24 24 24 24 24 24 24 24 24 $$$$$$$$$$$$$$$$
0038b0 24 24 24 24 00 00 9C 60 04 00 9F 60 00 00 AF 50 $$$$.....P
0038c0 38 74 FA 13 74 2E EB 18 74 74 65 73 74 03 00 B6 8t..t...ttest...
0038d0 38 00 00 00 00 00 03 00 17 00 00 00 00 00 01 04 00 8.....
0038e0 01 07 06 01 07 06 02 07 01 02 02 04 01 04 08 01 .....
0038f0 01 09 02 02 06 01 02 01 1B 43 44 4E .....CDN
3 byte(s) selected. Pos: 38F9h (14585) CAP OVR
```



Structural detection

- ▶ Method based on inspection of file properties/contents
 - ▶ E.g., content modifications, size difference, checksum / hash
- ▶ Analysis of modification of **file contents**:
 - ▶ If you have a copy of the original (virgin) file, it can be compared to the modified suspect/carrier file
 - ▶ Many tools can help view and compare the contents of hidden file to identify inconsistencies and patterns, e.g., Hex Editor
- ▶ Reviewing multiple files may identify a **signature** pattern related to the steganography program
 - ▶ Signature reveals program used to hide the message!
 - ▶ No original file necessary to compare it to



Structural detection: Properties and checksums

- ▶ Compare the properties of the files
 - ▶ 04/04/2003 05:25p 240,759 helmetprototype.jpg
 - ▶ 04/04/2003 05:26p 235,750 helmetprototype.jpg

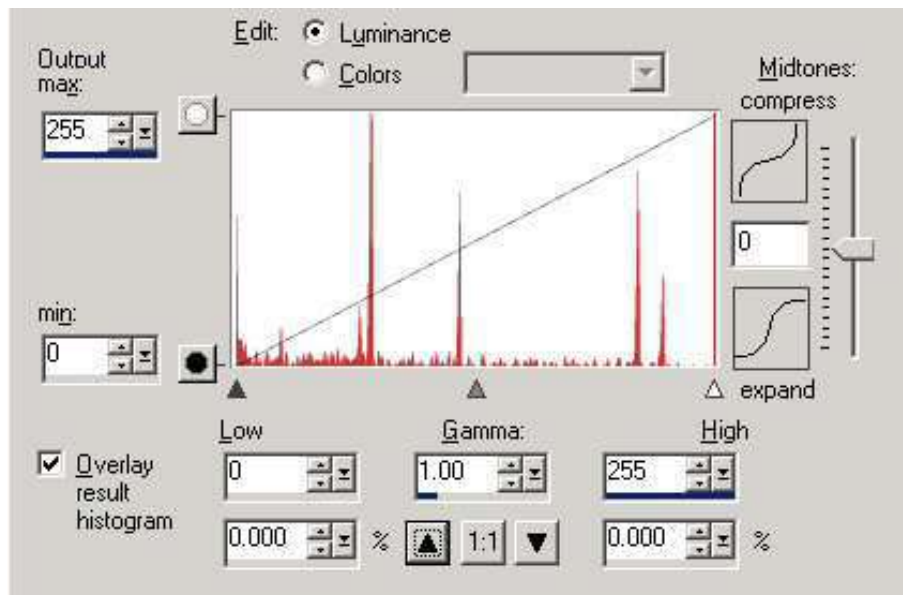
- ▶ Compare the checksum / hash of the files
 - ▶ C:\GNUTools>cksum a:\before\helmetprototype.jpg
3241690497 240759 a:\before\helmetprototype.jpg
 - ▶ C:\GNUTools>cksum a:\after\helmetprototype.jpg
3749290633 235750 a:\after\helmetprototype.jpg



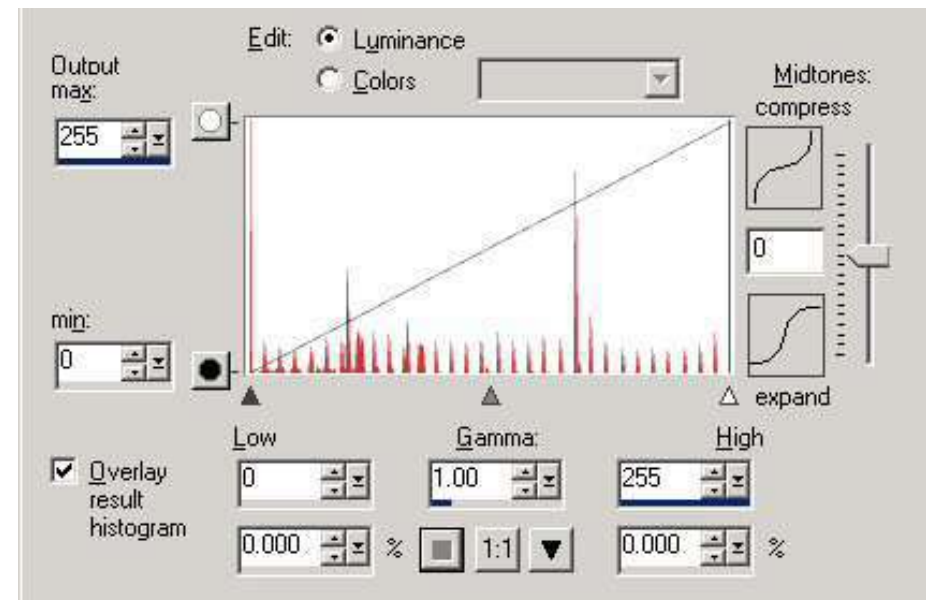


Statistical detection

- ▶ Example of histogram analysis
 - ▶ Can be used to possibly identify a file with a hidden message
 - ▶ This histogram (right) has a very noticeable repetitive trend



original



stego



Steganalysis of LSB embedding

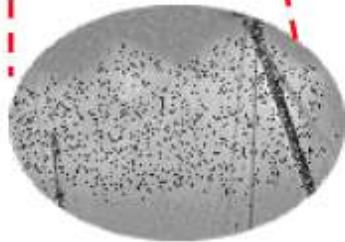
- ▶ In a “typical” natural scene, the number of even gray values is not the same as the number of odd values
- ▶ If you embed a 0-1 message string into the least significant bits (LSBs) of an image, then (since it is uniformly distributed), there will be approximately the same number of even and odd values
- ▶ Statistical “attack” to detect this anomaly
 - ▶ Histogram analysis (PoV analysis)
 - ▶ Regular-singular (RS) analysis
 - ▶ Sample pair (SP) analysis
 - ▶ LR Cube analysis
 - ▶ Etc.

Visual detection

- ▶ Detecting steganography by viewing it



original



**Simple observation
shows grainy noise
Introduced into
the steganogram**



carrier



steganogram



Least significant bits: black for LSB=0,
white for LSB=1

**Looking at LSB 1 layer seems
inconclusive**



Steganalysis attacks

- ▶ The steganalysis algorithm will depend on existing artifacts
 - ▶ **Stego-only attack:** Only the stego-object is available for stegoanalysis
 - ▶ **Known cover attack:** The original cover-object and stego-object are both available
 - ▶ **Known message attack:** Sometimes the hidden message may become known to the stegoanalyser. Analyzing the stego-object for patterns that correspond to the hidden message may be beneficial for future attacks against that system. (Even with the message, this may be very difficult and may even be considered equivalent to the stego-analysis)
 - ▶ **Chosen stego attack:** The stegoanalysis generates a stego-object from some steganography tool or algorithm from a chosen message. The goal in this attack is to determine corresponding patterns in the stego-object that may point to the use of specific steganography tools or algorithms
 - ▶ **Known stego attack:** The steganography algorithm is known and both the original and stego-objects are available



Types of steganalysis

▶ **Targeted steganalysis**

- ▶ Relies on knowing the method used to hide the data & using known distinguishing statistics to detect stego images
- ▶ Sometimes steganalysts reverse engineer steganographic methods

▶ **Blind steganalysis**

- ▶ Most beneficial to forensics because it's not based on knowing the algorithm
- ▶ Most difficult because the type of images and method of hiding data are enormous and continuously changing
- ▶ The current trend is to develop a neural network using training images and multiple statistical features

Watermarking



Cryptograpy & Steganography & Watermarking

- ▶ **Cryptography** is about protecting the content of messages (their meaning).
- ▶ **Steganography** is about concealing the existence of messages
- ▶ **Watermarking** is about establishing identity of information to prevent unauthorized use
 - ▶ They are imperceptible
 - ▶ They are inseparable from the works they are embedded in
 - ▶ They remain embedded in the work even during transformation



Watermarking examples

- Detect bill counterfeiting

Embedded
watermark





Examples: UV watermarking

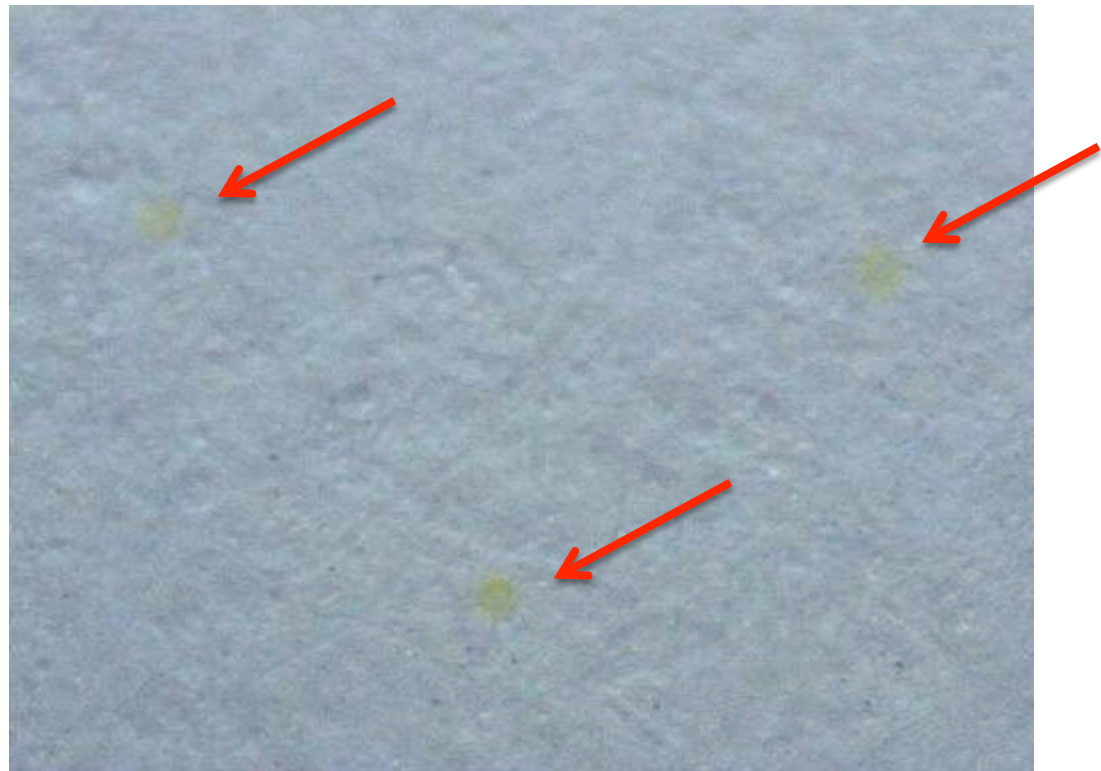
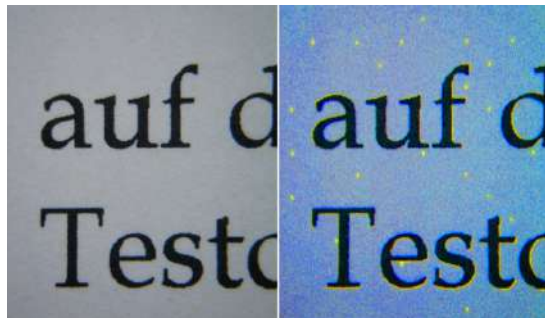
- ▶ Embedded watermark visible under UV light





Examples: Machine ID codes in laser printers

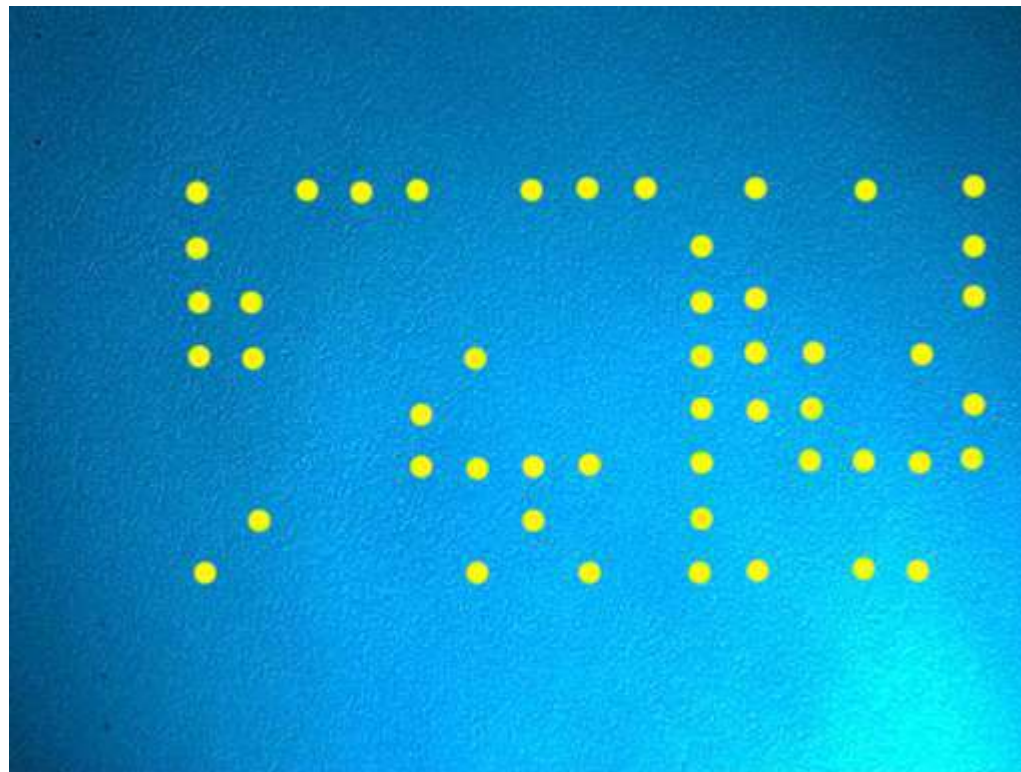
- ▶ Some printers print yellow tracking dots on their output
 - ▶ Printed in a regularly repeating pattern across the entire page





Examples: Machine ID codes in laser printers

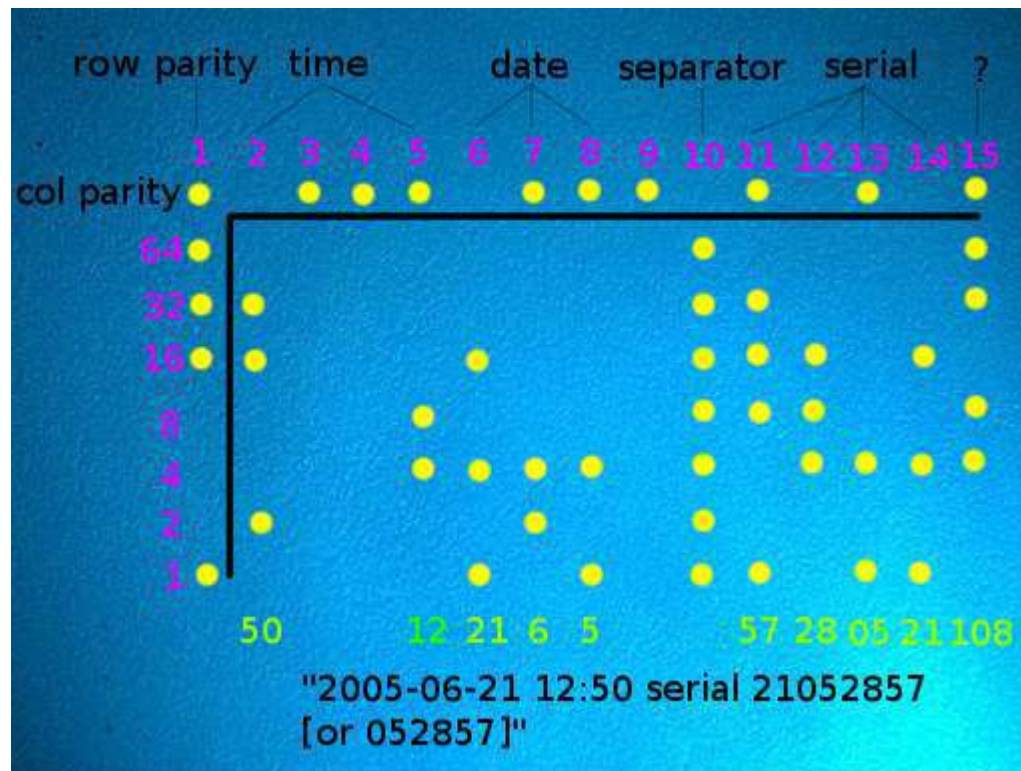
- ▶ Here, the dots are highlighted





Examples: Machine ID codes in laser printers

- By decoding the tracking dots, the ID can be recovered





Visible watermarks





Watermarking applications

- ▶ **Copyright protection**
 - ▶ Embed info about owner to prevent others from claiming copyright
 - ▶ Require very high level of robustness

- ▶ **Copy protection**
 - ▶ Embed watermark to disallow unauthorized copying of the cover
 - ▶ For example, a compliant DVD player will not playback or data that carry a "copy never" watermark

- ▶ **Content authentication**
 - ▶ Embed a watermark to detect modifications to the cover
 - ▶ The watermark in this case has low robustness, "fragile"



Steganography vs. Watermarking: Goals

- ▶ Both techniques **hide** a message m in some cover data d , to obtain d' , practically indistinguishable from d
- ▶ However, they have different goals:

Steganography

- ▶ An eavesdropper must not be able to **detect** the presence of m in d'
- ▶ Primarily for 1-to-1 communication

Watermarking

- ▶ An eavesdropper cannot remove or replace m in d'
- ▶ Primarily for 1-to-many communication



Steganography vs. Watermarking: Requirements

Steganography

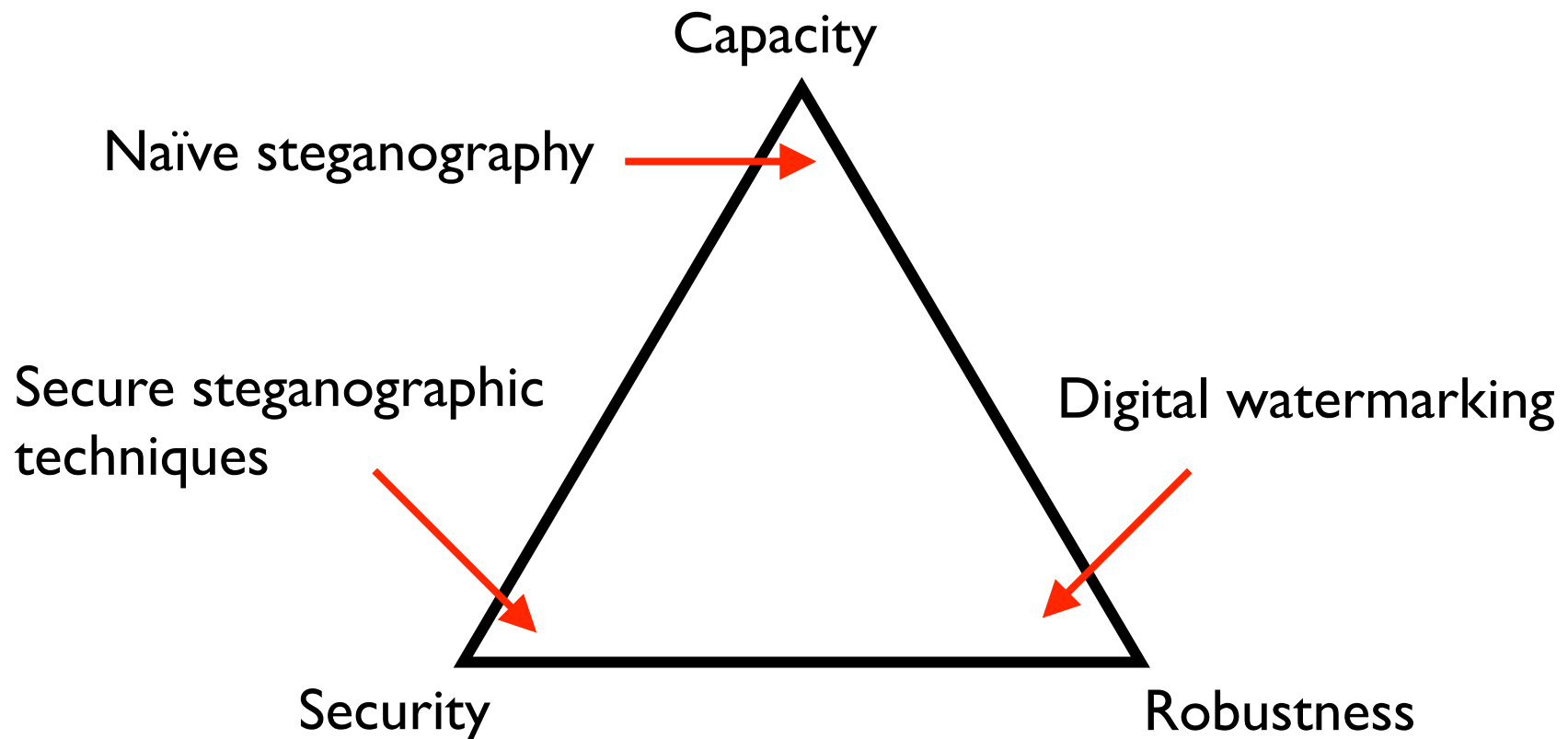
- ▶ Robustness not typically an issue
- ▶ Capacity desired for message is large
- ▶ Always invisible
- ▶ Typically dependent on file format

Watermarking

- ▶ Robustness of watermark is a main issue
- ▶ Known watermark may be there
- ▶ Can be visible or invisible
- ▶ Watermark can be considered to be an extended data attribute

The “magic” triangle

- ▶ Trade-off between capacity, security, and robustness

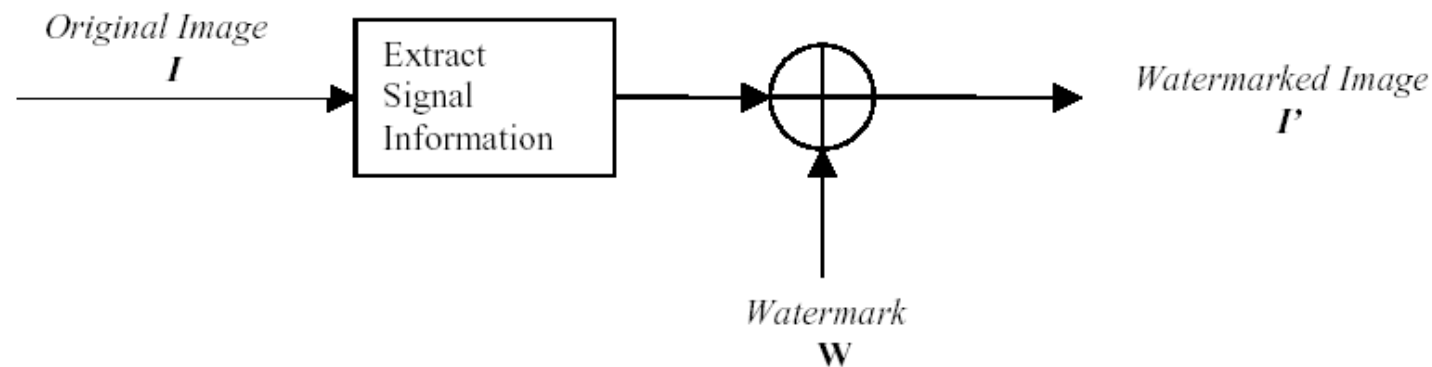




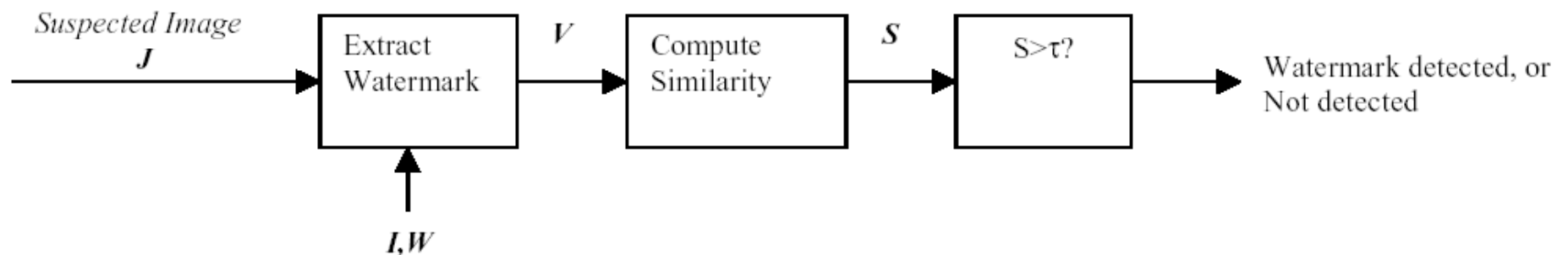
Digital watermark

- ▶ A digital signal or pattern inserted into a digital image

Watermark Transmission:



Watermark Detection:

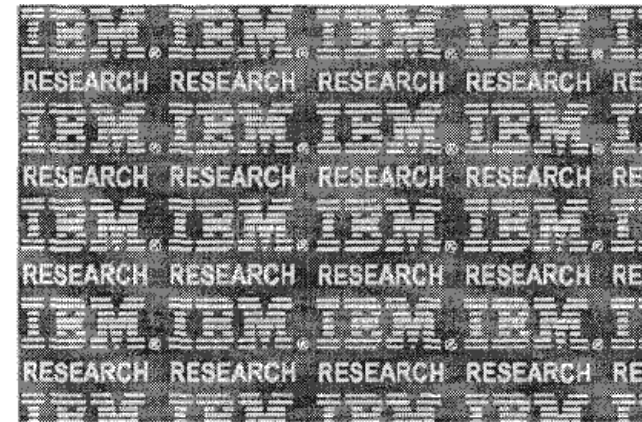




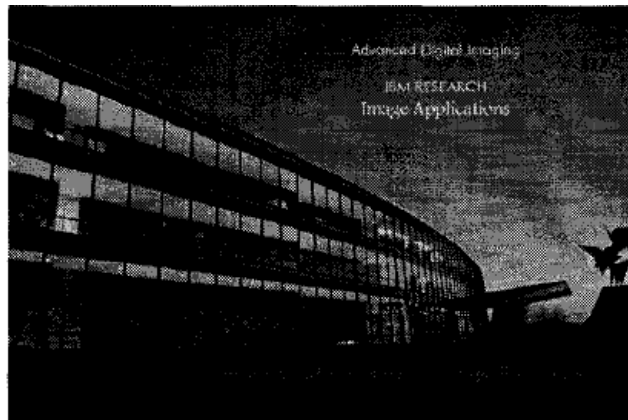
A simple technique: Checksum embedding



Original Image with watermark embedded



Watermark



Altered Image



Reconstructed Watermark



A simple technique: Checksum embedding

- ▶ Recover the watermark by applying a checksum function to each pixel of auth image and check LSBs

Corresponding pixels

$$f(\blacksquare) = 1$$



Perturb



Authenticated image



Binary logo



A simple technique: Checksum embedding

- ▶ Three key-dependent binary valued functions f_R, f_G, f_B

$$f_{R,G,B} : \{0, 1, \dots, 255\} \rightarrow \{0,1\},$$

- ▶ are used to encode a binary logo B . The values are perturbed in such a manner so that

$$B(i,j) = f_R(R(i,j)) \oplus f_G(G(i,j)) \oplus f_B(B(i,j)) \text{ for all } (i,j)$$

- ▶ The image authenticity is verified by checking the relationship

$$B(i,j) = f_R(R(i,j)) \oplus f_G(G(i,j)) \oplus f_B(B(i,j)) \text{ for each pixel } (i,j)$$

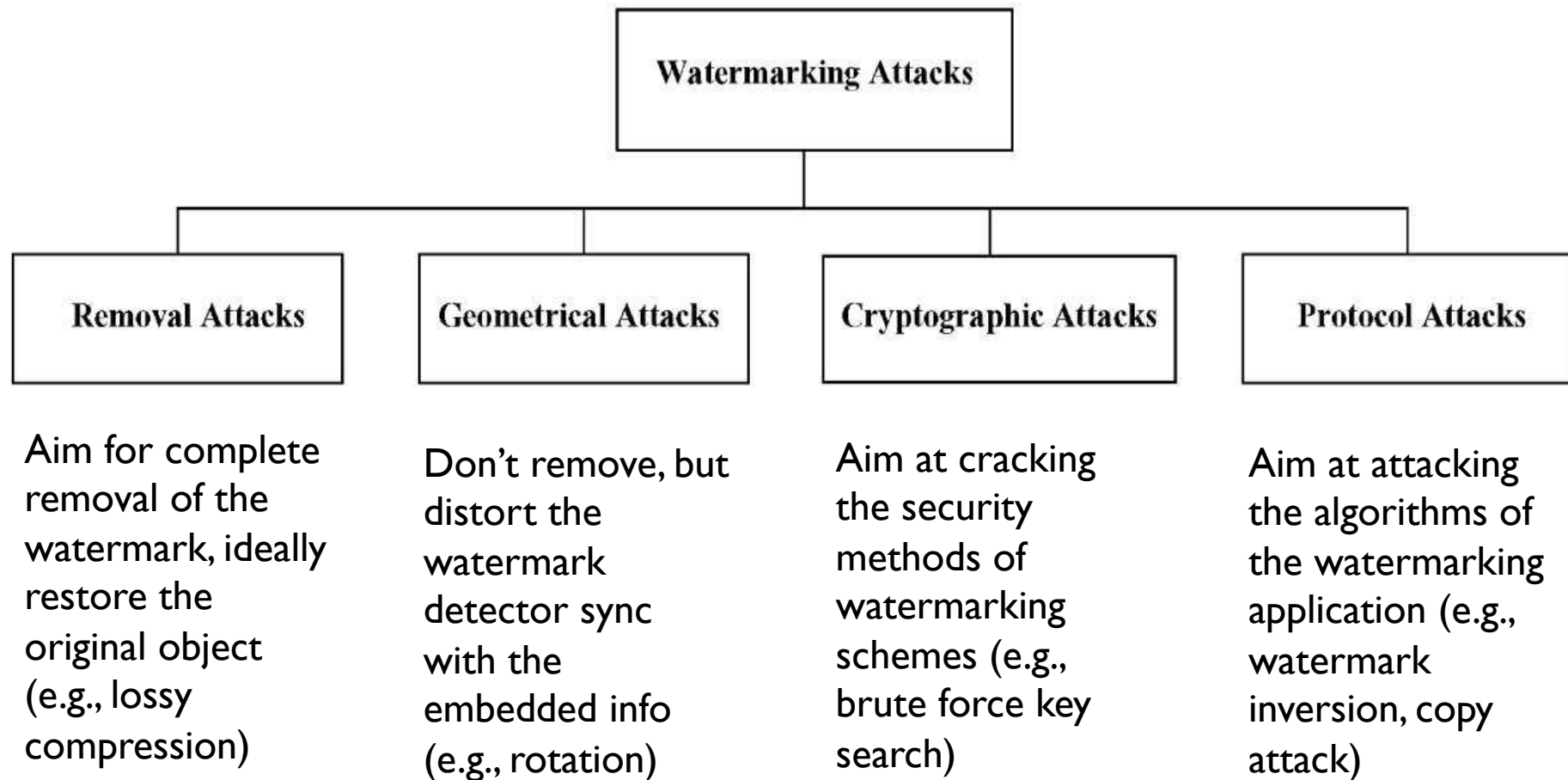


Attacking a watermarked image

- ▶ Three effects make detection of watermarking useless
 - 1. Watermark cannot be detected
 - 2. False watermarks are detected
 - 3. Unauthorized detection of watermark



Watermark attacking methods





Conclusions

- ▶ Digital steganography is an increasingly used technique for concealing communications within criminal activities and is difficult to mitigate by investigators
- ▶ On the other hand, digital watermarking helps investigators to trace the real identity of digital media
- ▶ Both fields are relatively young, and research is ongoing in order to increase the security and robustness of these techniques



References

► Primary bibliography

- Abbas Cheddad, Joan Condell, Kevin Curran and Paul Mc Kevitt. *Digital Image Steganography: Survey and Analysis of Current Methods*. Signal Processing, Volume 90, Issue 3, March 2010



Next class

► II.3 Storage and Memory Analysis